

VITAL INTELLIGENCE

ASSIGNMENT 02: PRODUCT REQUIREMENT DOCUMENT

Why? (Objective)

User :

Health-conscious individuals seeking weight management and wellness struggle with existing calorie-tracking apps because they require extensive manual food logging, rely on inaccurate calorie estimates, and fail to provide personalized insights into individual metabolic responses to food.

For Business:

Consumers increasingly seek personalized nutrition guidance, yet existing calorie-tracking apps suffer from low long-term engagement due to manual logging fatigue and limited personalization. Investing in a solution that automates nutrition tracking and delivers individualized food-response insights can improve user retention, increase subscription value, and create strong retention.

Who are the users ?

Health conscious, wellness focused users who already use wearables, apps to track their calories, weight management focussed users

Persona :

Megan McCormack

38 year old Pharmacy Manager ,she wants to lose weight to avoid risk of Diabetes. Tried food diary apps and cannot keep up with logging food daily

Needs :

- Needs to seamlessly track her daily calories in, exercise, eating pattern and notification when she is over than required target
- Accurate estimate of her calorie needs in order to reach target weight
- Interested specifically in how her body responds to foods that she eats everyday
- Need to know which diet, habits works for her
- Needs alerts when there are any patterns (food/behaviour) that can increase her risk of diabetes

Problems we are solving :

Manual food logging feels unsustainable

No insights on food context with users biometric data

Frustrations expressed :

- " "I know I should track what I eat. But after a long day, I just can't face logging every meal."

Solution

Vital Intelligence concept - logging happens automatically, and each evening the user gets a plain-language picture of their day. Narrative provides them metabolic awareness,shows their patterns along with any action required

Product Flow

Flow Diagram

System Overview : How the Three Components Connect

Every eating event flows from hardware detection through on-device processing to the end-of-day output. The camera fires repeatedly across the eating window, not just once.



Camera

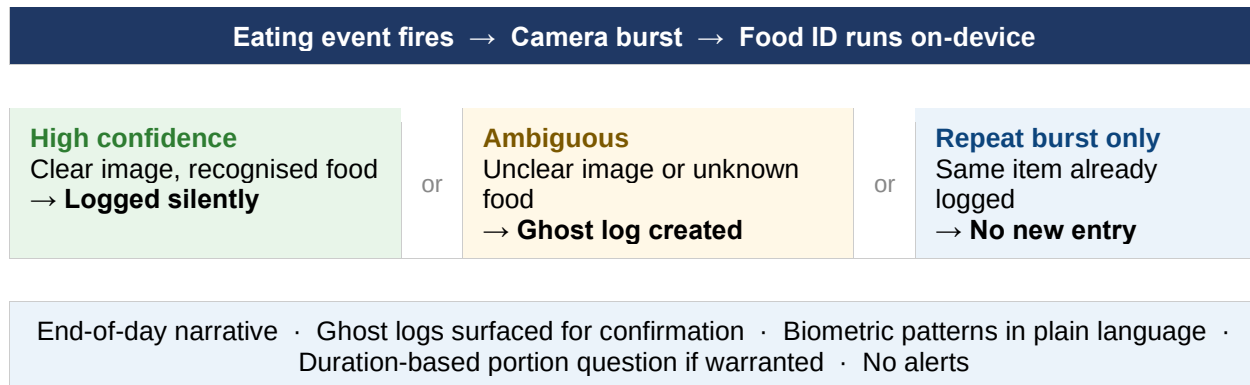
The camera is dormant by default. It wakes on two conditions only, and goes dormant again after each burst. The wearable monitors hand-to-mouth gesture activity continuously in the background and this is not a continuous camera activity.

Trigger	Behaviour
Eating event fires (initial)	Camera wakes. 4-frame burst over 3 seconds,sends food images to device and goes dormant. Food images analysed and logged automatically by VI.Biometric window opens post meal and metabolic analysis done.

Gestures continue or resume within 10 min	Camera wakes. New burst. Camera goes dormant. App compares new images against already-logged items in this window. New items added as ghost logs automatically.
15-minute continuous gesture-free gap	Window closes. Episode ends. Any gesture resuming after this gap = new separate eating event.

Food Logging - Full Decision Flow

Every image captured by the camera whether from the initial burst or a repeat burst → passes through the same confidence assessment. The outcome is either silent logging, a new ghost log, or a same-item match with no new entry.



Onboarding Timeline

Days 1–7 Learning	Days 7–14 Baseline forming	Day 14+ Patterns emerging	Ongoing Model matures
Events logged. Ghost logs confirmed at end of day. Meal duration baseline from stated preference. No metabolic insights yet.	Biometric curves and duration baselines built based on observed patterns.	Duration baseline personalised. Metabolic and behavioural patterns appear in narrative.	Ghost log accuracy improves. New-vs-old comparison becomes more reliable as item history builds.

Prototype

<https://vital-intelligence.ai.studio>

High Level Features :

1.As a user

When I have my meal or drink

I want 'Food logging' to happen automatically

So that I don't have to manually log them after every meal or remember to log them end of the day

2.As a user

After I have my meal or snack

I want to track my biometric response to the meal

So that I can understand if the food or habit is aligned to my health goals

How do we measure success?

METRIC TYPE	METRIC NAME	DEFINITION	WHY IT MATTERS
KEY METRIC	App Store Conversion Rate	(Total app downloads/total app store page views)*100	conversion rate between 25% and 30% indicates people are interested in the app
KEY METRIC	Rate of retention on day 14	% users who are actively using the app after 2 weeks of onboarding (users active on day 14 /total no of users) * 100	It measures how many users stick around after two weeks compared to those who drop off

KEY METRIC	Churn rate by day 14	percentage of users who abandoned the app or went inactive by week two.	A higher churn rate (40% or above) indicates users are not finding the auto logging or insights valuable
Supporting Metric	No of clicks for insights card	No of views for metabolic insights in a week	Indicates stronger user engagement as they view the insights card to check their narrative for the day
GUARDRAIL	Wearable Disconnection Rate	This measures the percentage of users who have either manually unlinked their wearable, revoked Bluetooth/API permissions, or whose devices have stopped syncing data to VI app for more than 48 hours.	VI core action is auto logging meal and delivering metabolic insights after meal. If the wearable disconnects, that automation breaks. The user is left with an empty dashboard, which immediately triggers churn.

Risks and Mitigation

ASSUMPTION AT RISK	FAILURE MODE	IMPACT	LIKELIHOOD	MITIGATION	DETECTION SIGNAL	CONTAINMENT
Eating event detection false positives will be minimal	System triggers false eating events	High	Medium	Establish a baseline of typical eating times during onboarding. Require user confirmation via push notifications for detected events during the first two weeks of onboarding.	Churn rates spike; high frequency of user-deleted or ignored meal logs.	Provide an immediate fallback option for users to manually scan or override the food entry.
Camera-captured images will be clear and reliable for	Poor lighting, bad angles, or blurry photos prevent the image	HIGH	MED	If a captured image is low quality, utilize "ghost logging" to run predictive	High rate of unrecognized or unclassified image uploads	Partner with or offer hardware recommendations for wearables that feature

food recognition.	recognition engine from identifying the food.			analysis on frequent meals and ask the user to confirm.	in the backend logs.	integrated, optimized cameras.
Connectivity and data synchronization between the device and the app will remain seamless	Bluetooth drops, background sync freezes, or network errors pause data transmission.	HIGH	MED	Implement local data caching on the device when a connection drops. Pause active insights and automatically run a gap-filling sync routine once the connection is re-established.	A drop in data payloads received over a rolling 12-to-24-hour window.	Ask the user to confirm the missing data block once sync recovers to reconcile the timeline.
The system can accurately capture and generate insights for all metabolic shifts.	Subtle or complex metabolic responses cannot be confidently isolated or interpreted by the algorithm during the initial weeks	HIGH	LOW	Filter out ambiguous or low-confidence data. Only provide high-confidence insights for distinct, clear meal responses while omitting the rest to maintain data integrity.	High user drop-off or low engagement on generalized or vague insight screens.	Restrict insight generation exclusively to definitive metabolic spikes, ensuring the user only receives actionable data.

Dependencies

1. Open Questions
2. Infrastructure requirements -
3. Budget Approvals -N/A
4. Partner Support- N/A
5. Internal Dependencies